

1 (a) Define gravitational potential at a point.

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..... [2]

(b) A uniform spherical planet has radius R . The gravitational potential at its surface is $-\Phi$.

P is a point on a line that passes through the centre of the planet, at variable displacement x from the centre, as shown in Figure 1.1.

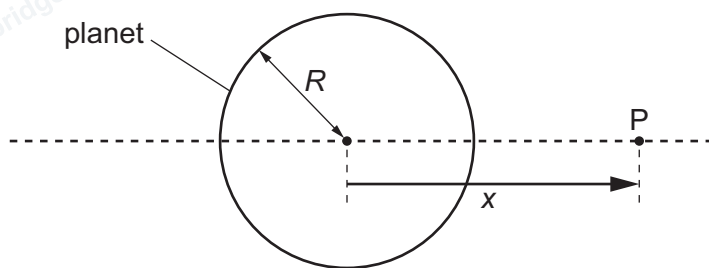


Figure 1.1

Figure 1.2 shows the variation with x of the gravitational potential ϕ due to the planet in the region outside the planet close to its surface.

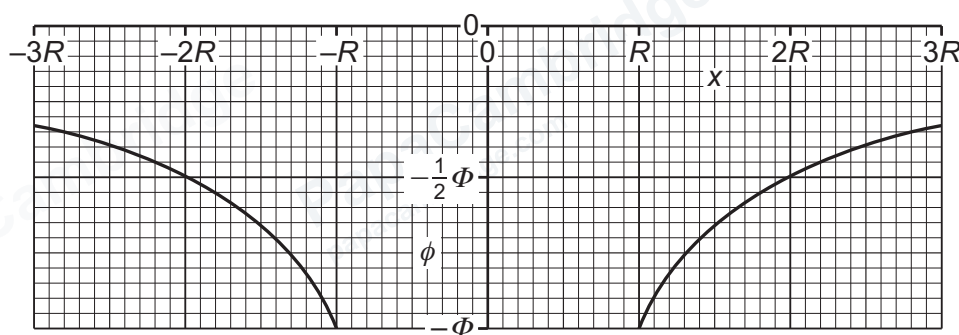


Figure 1.2

(i) Explain why ϕ is negative for all values of x .

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..... [2]

- (ii) State an expression, in terms of Φ and R , for the mass M of the planet. Identify any other symbols that you use.

$$M = \dots\dots\dots [2]$$

- (iii) Determine an expression, in terms of Φ and R , for the gravitational field strength g_0 at the surface of the planet.

$$g_0 = \dots\dots\dots [1]$$

- (iv) On Figure 1.3, sketch the variation of the gravitational field g with x between $x = -3R$ and $x = 3R$. Do **not** include the region inside the sphere between $x = -R$ and $x = R$.

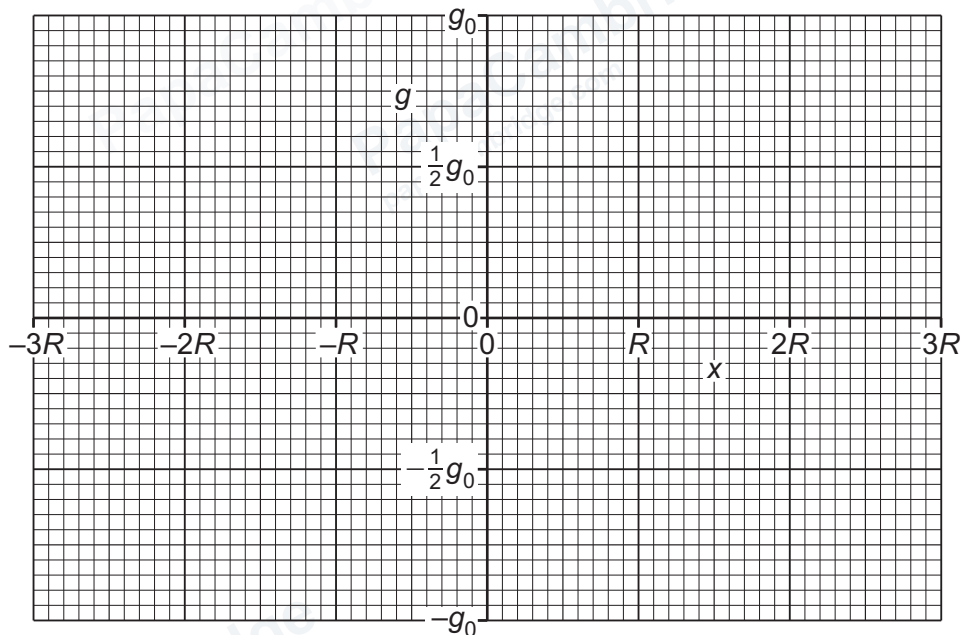


Figure 1.3

[3]

[Total: 10]